

CASABLANCA ANFA, Morocco

WMO: 601550

Lat: 33.557N

Lon: 7.660W

Elev: 203

StdP: 14.59

Time Zone: 1.00 (E01)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.3	46.2	37.2	32.8	51.1	39.8	36.3	52.0	16.8	57.6	14.2	58.3	3.1	200	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	85.1	70.8	81.7	71.4	79.4	71.2	75.2	80.2	74.0	78.7	72.9	77.5	6.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	125.8	78.1	72.2	119.9	76.6	71.6	117.4	75.9	38.8	80.3	37.7	78.8	36.7	77.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.4	12.2	11.0	DB	40.8	94.4	3.1	5.7	38.6	98.5	36.8	101.8	35.1	105.1	32.8	109.2	(4)
			WB	38.5	77.4	3.3	2.2	36.1	79.0	34.2	80.3	32.3	81.5	29.9	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	56.2	57.2	60.4	62.5	66.2	71.0	73.5	74.4	72.6	69.1	62.6	58.4	(6)
		DBStd	7.29	3.72	3.79	4.17	3.53	3.64	3.15	2.75	2.38	2.92	3.70	4.66	3.78	(7)
		HDD50	3	2	1	0	0	0	0	0	0	0	0	0	0	(8)
		HDD65	1078	272	220	155	89	27	0	0	0	6	101	208		(9)
		CDD50	5624	196	203	323	376	502	630	730	755	678	593	377	261	(10)
		CDD65	1224	1	1	13	16	64	181	265	290	228	134	28	3	(11)
		CDH74	3818	5	6	76	66	157	461	890	1060	675	331	82	9	(12)
		CDH80	518	1	1	21	14	32	91	137	102	51	52	15	1	(13)
(14)	Wind	WSAvg	5.0	4.1	4.8	5.3	5.7	5.8	5.9	5.7	5.1	4.8	4.6	4.3	4.5	(14)
(15)	Precipitation	PrecAvg	16.6	2.6	2.1	1.9	1.5	0.7	0.2	0.0	0.3	1.6	2.8	3.0		(15)
(16)		PrecMax	31.7	12.4	8.4	7.4	6.1	3.9	1.0	0.3	0.2	1.8	5.2	10.0	14.9	(16)
(17)		PrecMin	5.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		(17)
(18)		PrecStd	5.6	2.4	1.8	1.6	1.1	0.9	0.2	0.1	0.0	0.4	1.5	2.1	2.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	71.8	73.6	81.9	81.8	85.5	92.3	91.6	87.6	84.9	85.8	82.1	73.8		(19)
(20)		0.4% MCWB	59.9	61.5	64.7	64.6	67.3	72.2	74.2	74.0	74.2	65.1	66.5	60.5		(20)
(21)		2% DB	67.5	68.8	74.1	74.9	78.3	81.0	83.2	82.2	80.7	80.2	75.4	69.5		(21)
(22)		2% MCWB	59.4	61.0	62.2	64.0	66.7	71.2	72.9	73.7	72.3	68.0	65.7	61.7		(22)
(23)		5% DB	64.6	66.0	70.0	71.2	74.3	78.5	80.2	80.2	78.8	77.1	72.0	66.8		(23)
(24)		5% MCWB	59.0	60.1	61.8	63.2	66.3	70.4	72.7	73.4	71.2	69.0	64.9	60.7		(24)
(25)		10% DB	62.8	64.1	67.6	68.4	71.9	75.8	78.3	78.7	77.3	75.2	70.0	65.3		(25)
(26)		10% MCWB	58.0	59.0	61.2	62.0	65.4	69.4	71.8	72.2	70.6	68.4	63.9	59.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	63.3	64.8	66.5	68.1	70.3	74.6	76.3	76.9	76.3	73.0	70.7	65.6		(27)
(28)		0.4% MCDB	67.4	70.1	75.0	74.7	78.0	82.3	83.2	82.0	81.4	77.3	75.1	69.4		(28)
(29)		2% WB	61.2	62.4	64.4	65.6	68.6	72.7	74.6	75.4	74.1	71.6	68.0	63.5		(29)
(30)		2% MCDB	65.2	66.2	70.8	71.9	74.9	78.9	80.0	79.8	78.2	75.8	72.6	67.5		(30)
(31)		5% WB	59.8	60.9	63.1	64.1	67.3	71.4	73.4	74.2	72.8	70.5	66.2	62.0		(31)
(32)		5% MCDB	63.7	64.7	68.5	69.1	72.8	76.9	78.6	78.5	76.8	74.6	70.6	65.7		(32)
(33)		10% WB	58.6	59.6	61.8	63.0	66.2	70.2	72.4	73.0	71.8	69.4	64.6	60.6		(33)
(34)		10% MCDB	62.3	63.2	66.6	67.7	71.3	75.1	77.3	77.2	76.0	73.3	69.3	64.3		(34)
(35)	Mean Daily Temperature Range	MDBR	13.8	13.5	13.6	12.0	10.9	9.3	9.0	9.0	10.2	12.4	13.5	13.5		(35)
(36)		5% DB MCDBR	15.5	16.0	18.6	15.8	14.7	13.4	12.6	11.1	11.0	14.6	15.7	14.9		(36)
(37)		5% DB MCWBR	10.4	10.5	11.0	9.0	7.5	6.6	6.0	5.8	5.8	7.9	9.7	9.8		(37)
(38)		5% WB MCDBR	14.2	14.5	15.8	13.3	12.9	11.5	11.0	10.5	10.5	12.3	13.6	12.7		(38)
(39)		5% WB MCWBR	10.2	10.1	10.1	8.5	7.3	6.4	6.2	6.0	6.3	7.5	9.0	9.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.360	0.414	0.409	0.433	0.465	0.507	0.504	0.445	0.407	0.368	0.343		(40)
(41)		taud	2.495	2.407	2.228	2.276	2.225	2.136	2.002	2.031	2.226	2.351	2.443	2.494		(41)
(42)		Ebn at Noon	279	281	272	279	272	262	251	249	259	261	263	268		(42)
(43)		Edh at Noon	27	33	43	43	45	49	56	54	42	35	29	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	952	1225	1591	1950	2192	2367	2376	2151	1776	1327	1005	872		(44)
(45)		RadStd	61	106	108	93	116	95	57	61	70	88	70	55		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)		N/A	N/A	-1.55	+1.67	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page